

EMCH 260

SOLID MECHANICS

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PREVIOUS ON CHAPTER 1

External Loadings

**Mechanics
of Solids**

Deformation

Resultant
internal
forces;
Stress

???



Chapter 2.1

- ✓ Understand the concept of normal and shear strain
- ✓ Apply the concept to determine the strains for various types of problems

DISPLACEMENT AND DEFORMATION

- **Displacement in Statics**

- A vector that represents a movement of a point in a body (due to applied loads) with respect to some reference system of axes
- Translation and/or rotation
- Both shape and size of the body do not change

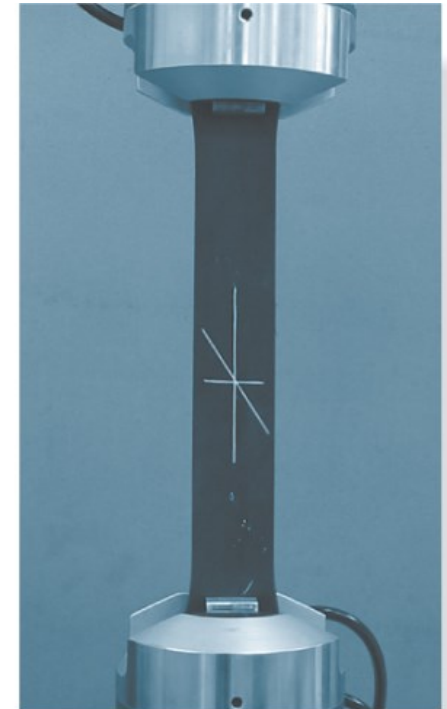
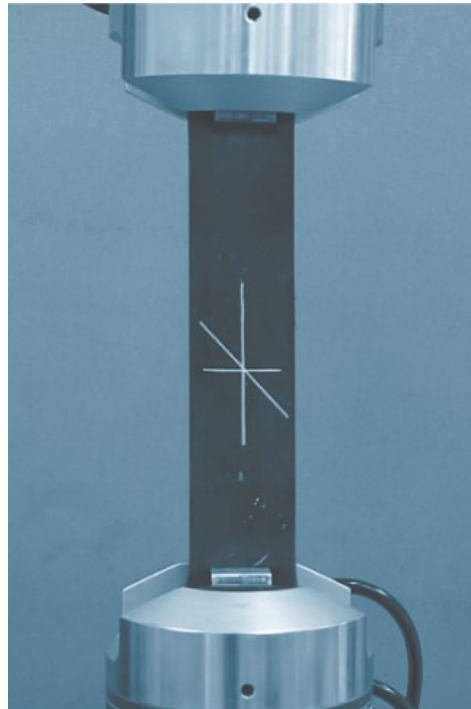
- **Deformation in Mechanics**

- A vector that represents a movement of a point in a body (due to applied loads) relative to another body point
- Both shape and size of the body could change (being deformed)
- Volume may be unchanged (special cases)

STRAIN

Intensity of deformation

Objects of the same materials but different sizes demonstrate different effects when subjected to the same load

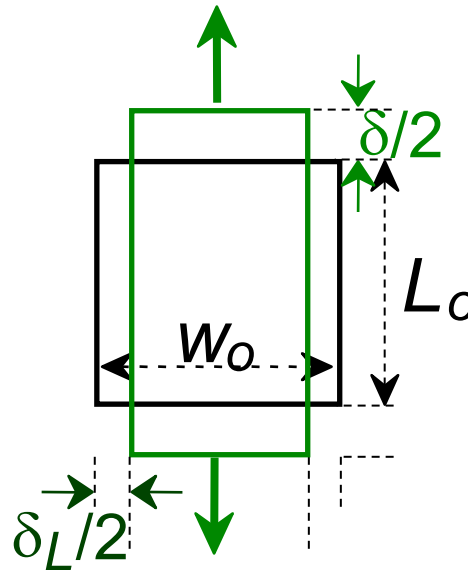


NORMAL STRAIN

Normal strain (ε): measures the change in size (elongation/contraction)

- **Tensile strain:**

$$\varepsilon = \frac{L - L_0}{L_0} = \frac{\delta}{L_0}$$



- **Lateral strain:**

$$\varepsilon_L = \frac{\delta_L}{W_0}$$

Positive ($\varepsilon > 0$): elongation

Negative ($\varepsilon < 0$): contraction

Units: dimensionless

SPATIAL DISTRIBUTION OF STRAIN

NOTE: under homogeneous and isotropic assumption, the normal strain at any point in the material is equal to the global (macroscopic) strain, i.e. *spatial distribution of strain is uniform and constant!*

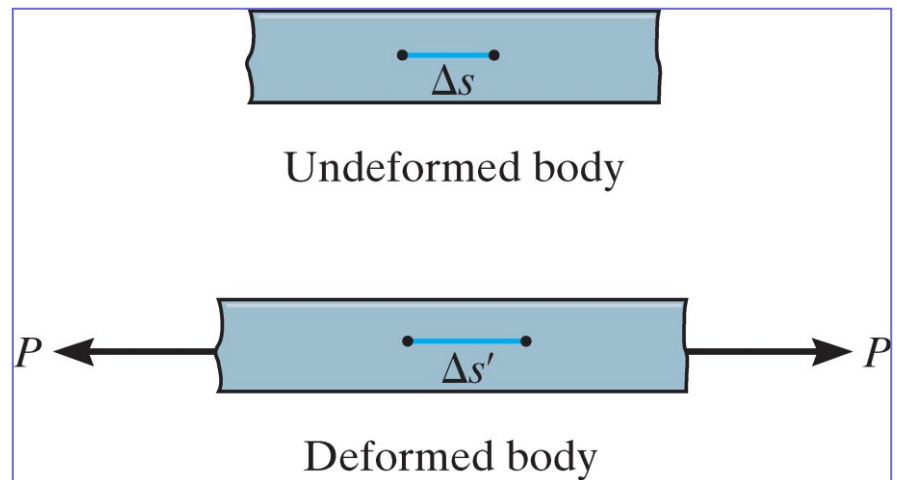
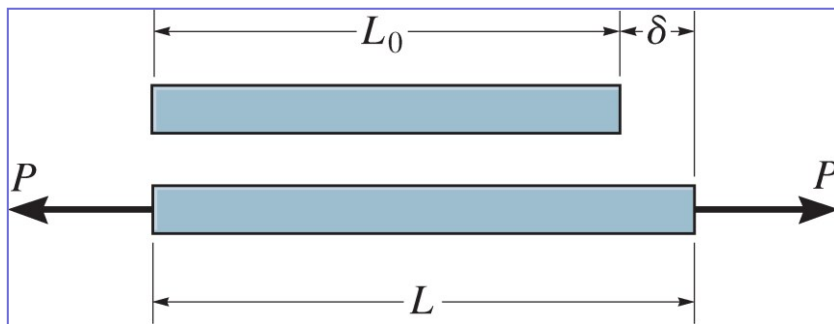
$$\varepsilon = \frac{\delta}{L_0} = \lim_{\Delta s \rightarrow 0} \frac{\Delta s' - \Delta s}{\Delta s}$$

Global strain

Local strain (strain at any point in the material)

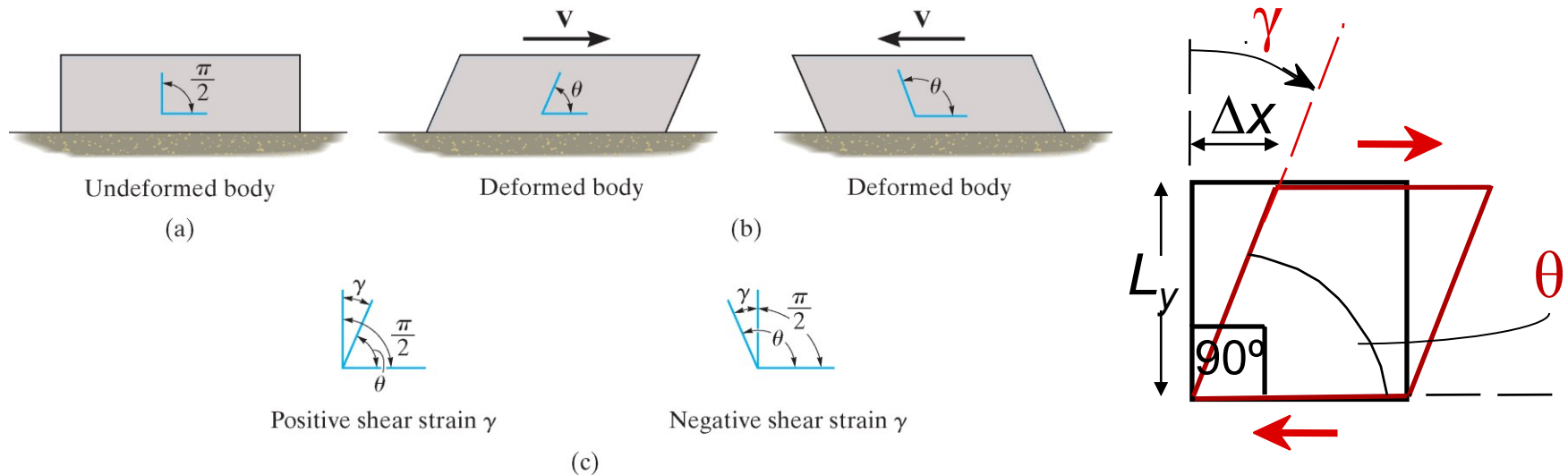
$$\Delta s' = (1 + \varepsilon)\Delta s$$

$$L = \int ds' = \int (1 + \varepsilon) ds$$



SHEAR STRAIN

Shearing strain (γ): measures the change in shape (angle formed by the sides of a body)



$$\gamma = \frac{\pi}{2} - \theta$$

Units: dimensionless

Small strain: $\varepsilon \ll 1$, $\gamma \approx \frac{\Delta x}{L_y} \equiv \tan \gamma$ ($\sin \gamma \approx \gamma$, $\cos \gamma \approx 1$)

REVIEW: Taylor Series Expansion

The Taylor series of a real or complex-valued function $f(x)$ that is infinitely differentiable at a real or complex number a is the power series

$$f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \dots,$$

Therefore, for small shear strain ($\gamma \rightarrow 0$), the Taylor series expansion of $\sin \gamma$ and $\cos \gamma$ at $\gamma = 0$ can be written as

$$\sin \gamma = [\sin \gamma]_{\gamma=0} + \left[\frac{\sin' \gamma}{1!} \right]_{\gamma=0} \cdot (\gamma - 0) + \left[\frac{\sin'' \gamma}{2!} \right]_{\gamma=0} \cdot (\gamma - 0)^2 + \left[\frac{\sin''' \gamma}{3!} \right]_{\gamma=0} \cdot (\gamma - 0)^3 + \dots$$

$$\sin \gamma = \gamma - \frac{\gamma^3}{3!} + \frac{\gamma^5}{5!} - \frac{\gamma^7}{7!} + \dots \quad \text{Taking first order,} \quad \sin \gamma = \gamma$$

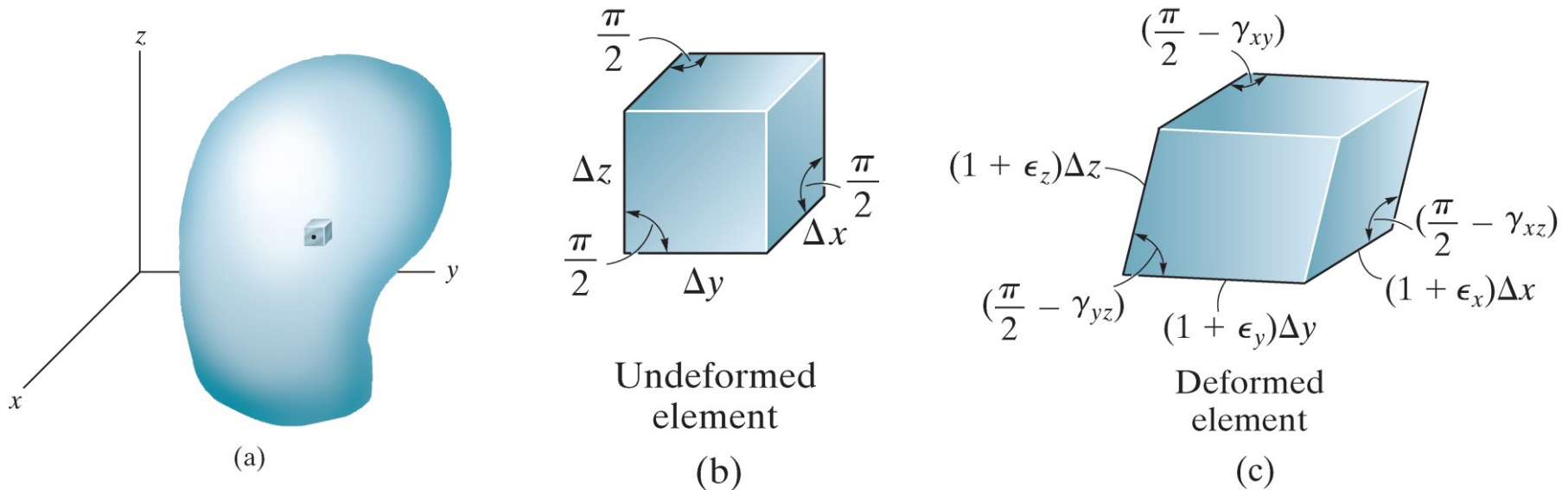
$$\cos \gamma = [\cos \gamma]_{\gamma=0} + \left[\frac{\cos' \gamma}{1!} \right]_{\gamma=0} \cdot (\gamma - 0) + \left[\frac{\cos'' \gamma}{2!} \right]_{\gamma=0} \cdot (\gamma - 0)^2 + \left[\frac{\cos''' \gamma}{3!} \right]_{\gamma=0} \cdot (\gamma - 0)^3 + \dots$$

$$\cos \gamma = 1 - \frac{\gamma^2}{2!} + \frac{\gamma^4}{4!} - \frac{\gamma^6}{6!} + \dots \quad \text{Again, taking first order,} \quad \cos \gamma \approx 1$$

$$\tan \gamma = \gamma + \frac{\gamma^3}{3} + \frac{2\gamma^5}{15} + \dots \quad \text{Again, taking first order,} \quad \tan \gamma \approx \gamma$$

CARTESIAN STRAIN COMPONENTS

The deformation of any cubic element in the material can be described using normal and shear strains.



New side lengths of cubic element:

$$(1 + \epsilon_x) \cdot \Delta x \quad , \quad (1 + \epsilon_y) \cdot \Delta y \quad , \quad (1 + \epsilon_z) \cdot \Delta z$$

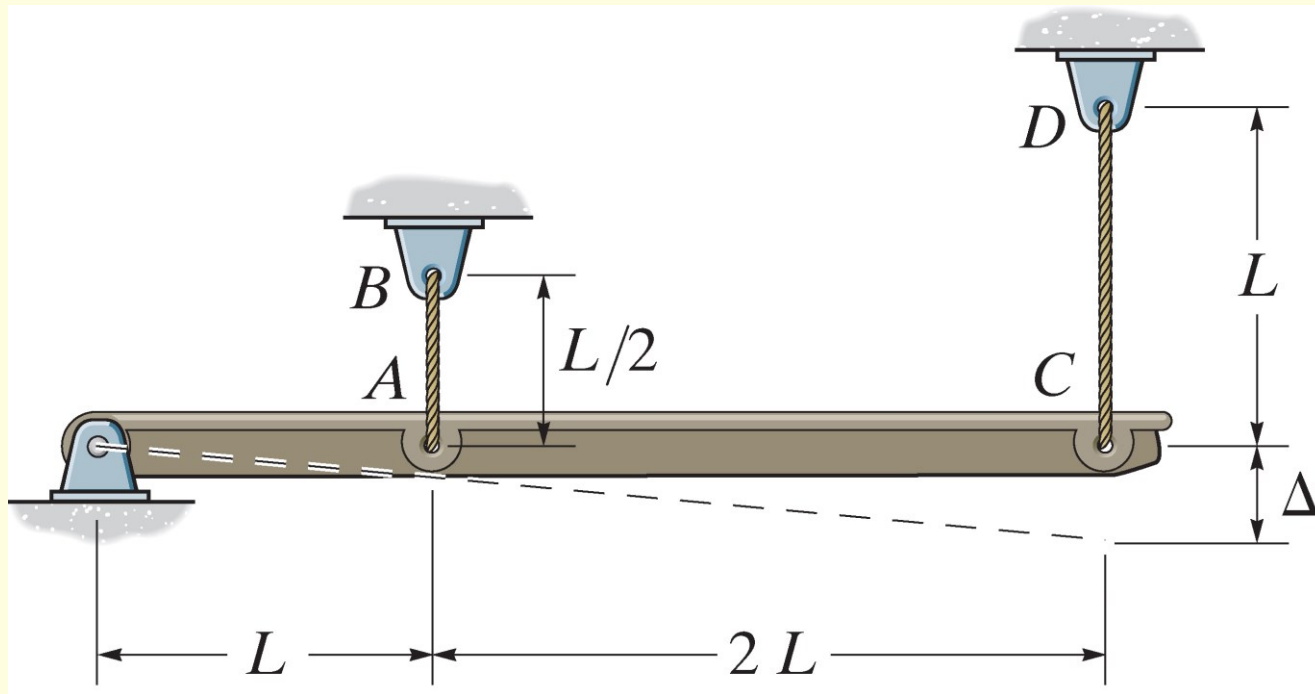
New angle between the sides:

$$\frac{\pi}{2} - \gamma_{xy} \quad , \quad \frac{\pi}{2} - \gamma_{yz} \quad , \quad \frac{\pi}{2} - \gamma_{xz}$$

The **normal strains** cause a **change in volume** of rectangular element, whereas the **shear strain** cause a **change in shape**.

IN-CLASS EXAMPLE 1

A loading causes the member to deform into the dashed shape. Determine the normal strains in the rope AB and CD .



IN-CLASS EXAMPLE 1 (cont'd)

Solutions

The normal strain in the rope AB (ϵ_{AB}) and CD (ϵ_{CD}) is determined by the elongation Δ' and Δ , respectively.

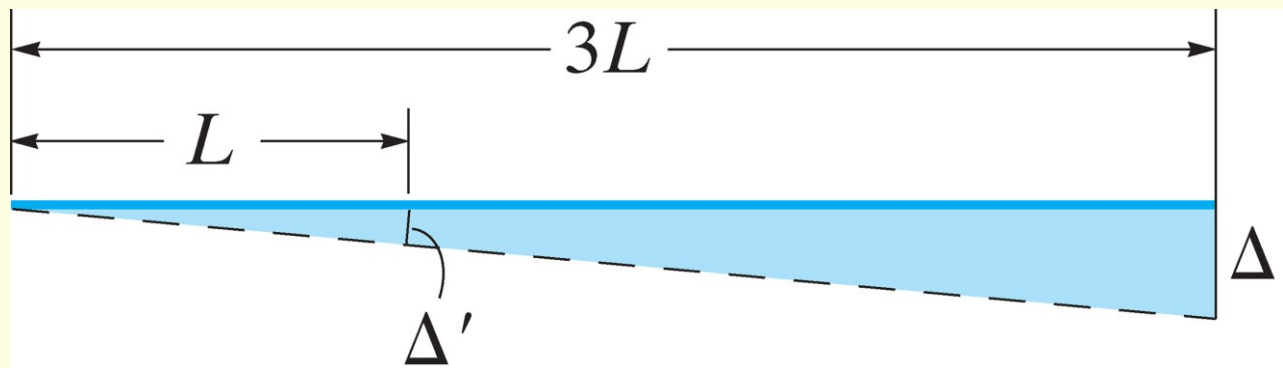
According to the geometry relationship, we have

$$\frac{\Delta'}{L} = \frac{\Delta}{3L} \rightarrow \Delta' = \Delta/3$$

Then,

$$\epsilon_{AB} = \frac{\Delta'}{L/2} = \frac{2\Delta}{3L}$$

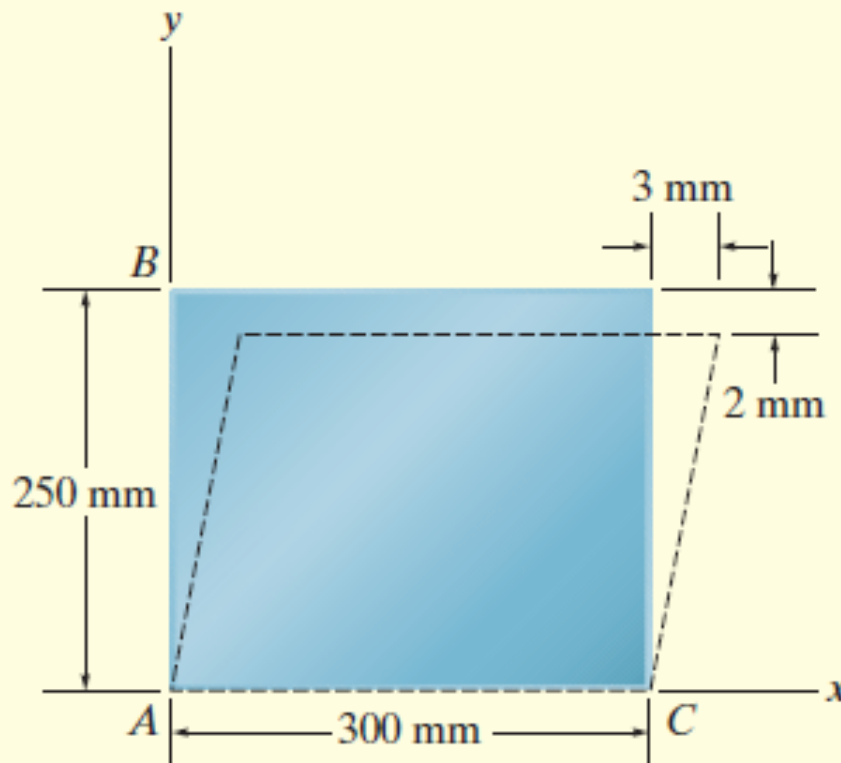
$$\epsilon_{CD} = \frac{\Delta}{L}$$



IN-CLASS EXAMPLE 2

Due to a loading, the plate is deformed into the dashed shape. Determine:

- (a) the average normal strain along the side AB
- (b) the average shear strain in the plate at A relative to the y axes.



IMPORTANT NOTE:

To calculate normal or shear strain, keep in mind the geometry (length or angle) *before* and *after* deformation

IN-CLASS EXAMPLE 2 (cont'd)

Solutions

(a)

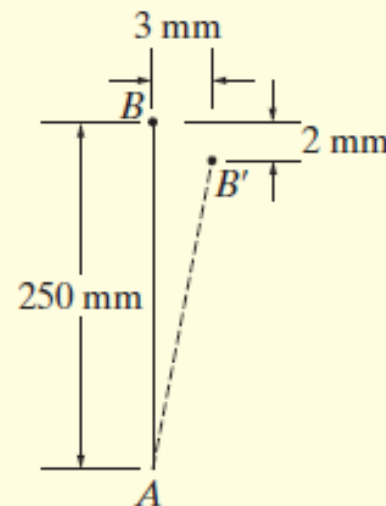
- Line AB , coincident with the y axis, becomes line after deformation, thus the length of this line is

$$AB' = \sqrt{(250 - 2)^2 + 3^2} = 248.018 \text{ mm}$$

- The average normal strain for AB is therefore

$$(\epsilon_{AB})_{avg} = \frac{AB' - AB}{AB} = \frac{248.018 - 250}{250} = -0.007928 \text{ mm/mm (Ans.)}$$

- The negative sign indicates the strain causes a contraction of AB .



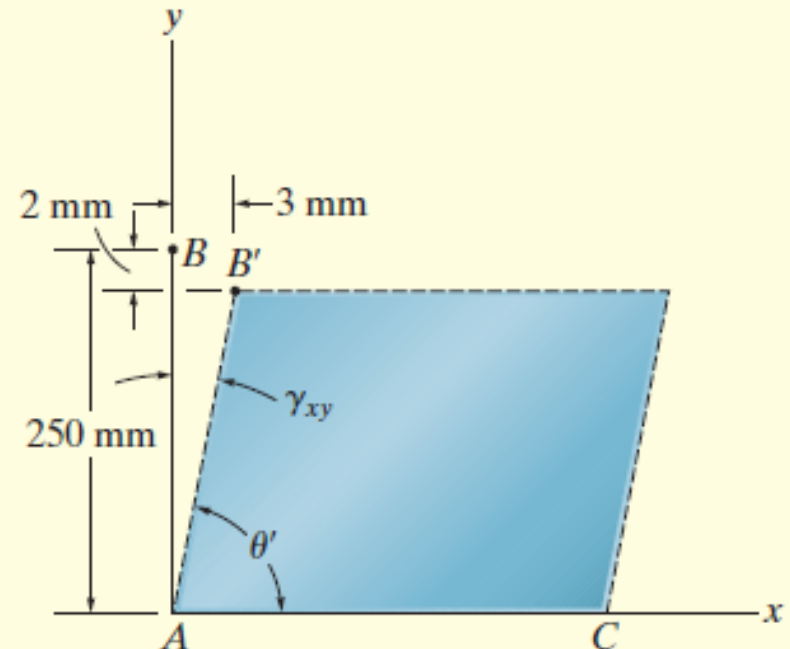
IN-CLASS EXAMPLE 2 (cont'd)

Solutions

(b)

- As noted, the 90 angle BAC between the sides of the plate, referenced from the x , y axes, changes to θ' due to the displacement of B to B' .
- Since $\gamma_{xy} = \frac{\pi}{2} - \theta'$ then γ_{xy} is the angle shown in the figure.
- Thus,

$$\gamma_{xy} = \tan^{-1}\left(\frac{3}{250-2}\right) = 0.121 \text{ rad (Ans)}$$



RELEVANT PRACTICE

- Assignment 1, problem #4 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- Optional Extra Exercise 1, problem #3 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13th 2021**)
- *Submit your solutions directly onto Blackboard*

OUT-OF-CLASS READING

- **Textbook, pp. 69 – 75 (in particular examples 2.1 – 2.3)**